

QUANTITATIVE TRADING IN STOCK USING DEEP EVOLUTION STRATEGY

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ABSTRACT

*This project aims to optimize a trading model for quantitative stock trading using **Deep Evolution Strategy (DES)**. Quantitative trading employs mathematical models and algorithms to analyze historical data and automate trading decisions. The model consists of a three-layer neural network: the input layer processes price changes over the past 30 days, the hidden layer with 500 neurons learns data patterns, and the output layer predicts three possible actions (hold, buy, or sell). DES trains the model through the following steps: random weight initialization, generating candidate solutions via perturbations, evaluating candidates based on return on investment, and updating weights according to rewards. The model's performance is assessed using a reward function that calculates the percentage growth of capital after simulated trading, allowing the model to refine its strategy.*

Results demonstrate that DES effectively optimizes non-gradient problems and learns optimal strategies automatically. Future improvements could incorporate technical indicators and simulate transaction costs to enhance realism and accuracy. This method combines data-driven analysis with flexible evolutionary optimization, showcasing its potential in quantitative trading applications.

Keywords: Deep Evolution Strategy, Quantitative Trading, Non-Gradient Optimization

1. INTRODUCTION

Quantitative trading leverages mathematical models, algorithms, and data analysis to identify market opportunities and automate trading decisions. Unlike traditional trading, which often relies on intuition, quantitative trading processes large datasets objectively, minimizing human error and optimizing performance.

This study focuses on using Deep Evolution Strategy (DES) to optimize a neural network model for automated stock trading. DES, inspired by natural evolution, is suitable for non-convex problems and eliminates the need for gradient-based optimization. By analyzing historical stock price data, the model learns to predict trends and make trading decisions—hold, buy, or sell—with the aim of maximizing returns.

2. METHOD

The proposed method utilizes a neural network model consisting of three layers: an input layer, a hidden layer, and an output layer. The input layer processes stock price changes over a 30-day window to generate features. The hidden layer, with 500 neurons, extracts patterns from the input data, while the output layer predicts one of three possible trading actions: hold, buy, or sell.

The model is trained using Deep Evolution Strategy (DES), an optimization method inspired by natural evolution. Instead of relying on gradient-based optimization, DES generates candidate solutions by adding random noise to the current model weights, creating a population of potential solutions. Each candidate is evaluated through a simulated trading process, and its performance is measured using a reward function, which calculates the percentage change in capital returns.

The weights of the model are updated based on the performance of high-reward candidates, guiding the model toward better trading strategies. By iteratively generating, evaluating, and refining the weights, DES enables the model to learn and adapt to market conditions, improving its ability to make profitable decisions.

3. CONCLUSION

This study demonstrates the potential of Deep Evolution Strategy (DES) in optimizing neural networks for quantitative stock trading. DES effectively handles non-convex problems, automates strategy refinement, and improves model performance without requiring gradient calculations.

While the results are promising, future work could incorporate technical indicators and transaction costs for greater realism and accuracy. Overall, this approach showcases DES as a powerful tool for advancing quantitative trading systems.